

TS-03 Dock Door Sensor Fault Isolation Guide

Category: Troubleshooting Guides

Synthetic sample document for public portfolio review.

DOCUMENT CONTROL

Guide ID: TS-03-DDR-330

Revision: Rev 1.0

Owner: Service Desk and Site Support

Use Case: Frontline diagnosis before depot repair or infrastructure escalation

Review Cycle: Every 6 months or after repeat incidents

OVERVIEW

This guide supports first-response diagnosis of fictional dock door photo-eye faults, false obstruction alarms, and intermittent beam-loss events. It is intended for site support, maintenance leads, and supervisors deciding whether a door can remain in limited service.

COMMON SYMPTOMS

- Door reverses immediately after closing even though the opening is clear.
- Controller panel shows intermittent obstruction or beam-loss alarms that clear without manual reset.
- Fault appears during sunrise or late-day glare on one door orientation only.
- Door closes normally in manual hold-to-run mode but faults in automatic mode.

LIKELY CAUSES

- Dirty or misaligned photo-eye lenses.
- Cable strain or intermittent break near the moving side of the frame.
- Sun glare or reflective shrink wrap causing false sensor readings.
- Controller input instability or failing sensor power supply.

DIAGNOSTIC WORKFLOW

- Secure the lane and confirm no trailer movement, pedestrian activity, or forklift traffic will intersect the door while testing is in progress.
- Inspect both lenses, brackets, and cable routing. Clean lenses before changing alignment.
- Observe controller input lights while gently moving the cable near the travel point to identify intermittent breaks.
- Test the door in manual mode and automatic mode to separate a control logic issue from a basic obstruction reading.
- If glare is suspected, shade the receiver temporarily and note whether the fault clears under the same operating cycle.

RECOVERY ACTIONS

- Realign brackets and retighten hardware if the beam path has shifted.

- Replace damaged cable sections or sensor heads if intermittent input changes are confirmed.
- Apply the approved glare shield or timing change only if the glare condition is documented and recurring on that door orientation.
- If the controller input remains unstable after sensor replacement, place the door out of service and escalate to controls support.

ESCALATION CRITERIA

- Escalate immediately if the door cannot be operated safely in either manual or automatic mode.
- Escalate to controls support if input lights disagree with physical sensor state after lens cleaning and cable inspection.
- Do not return the door to automatic service until three consecutive open-close cycles complete without false obstruction alarm.

FIELD NOTES

The distinction between glare and wiring faults creates a good retrieval target because both can look intermittent. The guide expects the reviewer to use controller input lights and temporary shading to separate them.

A later RAG evaluation could ask what evidence is needed before applying a glare shield. The answer should come from the diagnostic workflow and recovery sections together, not from a single sentence.

APPENDIX 1: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 2: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 3: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 4: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 5: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 6: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 7: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

APPENDIX 8: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.